



Peter Willendrup

Samples in McStas - an overview

Agenda



- General concepts (reminder)
- Overview of samples in McStas
- Focus on diffraction-oriented samples:
 - Incoherent scatterer
 - Powder
 - Single Crystal
 - File formats
- MDANSE2018 PDF's added with SANS and inelastic

Sample components in McStas v.2.5



Sample Process	Coherent Elastic	Inelastic	Incoherent Elastic	Inelastic	Absorption	Multiple Scattering
Incoherent	No	No	Yes	No	Yes	Yes
PowderN	Yes	No	Yes	No	Yes	Yes
Single_crystal	Yes	No	Yes	No	Yes	Yes
Isotropic_Sqw	Yes	Yes	Yes	Yes	Yes	Yes
Sans_spheres	Yes	No	Yes	No	Yes	No
SasView_models	Yes	No	Yes	No	Yes	No
Phonon_simple	No	Yes	No	No	No	No
Magnon_bcc	No	Yes	No	No	No	No
Single_crystal_inelastic	Yes	Yes	Yes	No	Yes	Yes

McStas sample overview table



McStas sample model functionality-matrix - McStas - ESS Confluence - Mozilla Firefox

McStas sample model functionality-matrix

Created by Peter Willendrup, last modified on Jun 19, 2018

(Master version available at the URL <https://confluence.esss.lu.se/display/MCSTAS/McStas+sample+model+functionality-matrix>)

Status of the McStas sample components, relevant for McStas 2.4.1 (Released summer 2017)

See also: [McStas sample models for Diffraction](#), [McStas sample models for Imaging](#), [McStas sample models for Large-scale Structures](#), [McStas sample models for Spectroscopy](#)

McStas sample comp + author info in <i>italic</i>	Model description	Main use areas	Incoherent scattering	Absorption	Bragg or other elastic scattering (type)	Inelastic scattering (type)	Multiple scattering	Non-trivial sample geometry
1 <i>Incoherent</i> (Vanadium, Plexiglass etc.) <i>McStas team</i>	Simple Incoherent scatterer	Generic, imaging	✓	✓	✗	✗	✓	✓
2 <i>Tunelling_sample</i> <i>McStas team / Kim Lefmann</i>	Idem 1, plus tunnelling peaks and QE broadening	Quasi-elastic scattering, backscattering	✓	✓	✗	✗ ✓ (Quasilelastic broadening + tunnel peaks)	✓ (analytic approach)	✓
3 <i>PowderN</i> <i>McStas team / Peter Willendrup</i>	Debye-scherrer cones, tabular input (<i>lau</i> / <i>laz</i>)	Powder diffraction , imaging	✓	✓	✓ (Debye-Scherrer cones)	✗	✗	✓
4 <i>Sample_nxs</i> <i>Mirko Boin, HZB</i>	Debye-scherrer cones, unit-cell / atom input list	Powder diffraction , (future: imaging)	✓	✓	✓ (Debye-Scherrer cones)	✗ ✓	✓	✗
5 <i>Single_crystal</i> <i>McStas team</i>	Bragg spots, tabular input (<i>lau</i>). "Perfect imperfect" single crystal with mosaicity / lattice variation	Single crystal and MX diffraction	✓	✓	✓ (Bragg spots)	✗	✓	✓
6 <i>Sans_spheres</i> (and other similar) <i>McStas team and Martin Cramer Pedersen, KU</i>	Hard spheres in thin solution and other models, defined per-component...	SANS	✓	✓	✓ - SANS	✗	✗	✗
7 <i>SANS_benchmark2</i> (and a few other stand-alone models) <i>Heinrich Frielinghaus, FZJ/JCNS</i>	Experimentally-benchmarked model set for SANS	SANS	✓	✓	✓ - SANS	✗	✓ up to 10 orders	✗
8 <i>SASview_models</i> (yet unreleased) <i>McStas team</i>	"Any" model from SASview / SASmodels	SANS	✓	✓	✓ - SANS	✗	✗ at this point	✗

McStas sample table overview



McStas sample model functionality-matrix - McStas - ESS Confluence - Mozilla Firefox									
https://confluence.esss.lu.se/display/MCSTAS/McStas+sample+model+functionality-matrix									
McStas sample comp + author info in italic	Model description	Main use areas	Incoherent scattering	Absorption	Bragg or other elastic scattering (type)	Inelastic scattering (type)	Multiple scattering	Non-trivial sample geometry	
<i>McStas team</i>									
9 <i>Multilayer_sample</i> <i>Rob Dalglish, ISIS STFC</i>	Multilayer-sample (dynamic scattering theory) with incoherent background	Reflectometry	✓	✓	✓ - Reflectivity curve	✗	✗	✗	
10 <i>Phonon_simple</i> <i>McStas team / Kim Lefmann</i>	Single-branch acoustic phonon in FCC lattice	Inelastic scattering phonons	✗	✗	✗	✓ (phonon, at this point FCC lattice only)	✗	✗	
11 <i>Isotropic_Sqw</i> <i>McStas team / Emmanuel Farhi</i>	Structure and dynamics in isotropic materials (liquids, powders etc.)	Inelastic scattering, diffraction, isotropic materials, imaging	✓	✓	✓ (Debye-Scherrer cones)	✓ isotropic inelastic scattering	✓	✓	
12 <i>Res_sample</i> <i>McStas team</i>	Resolution-oriented sample component	Generic	✓	✗	✗	✓ flat, isotropic inelastic scattering	✗	✗	
13 <i>TOFRes_sample</i>	Idem Res_sample, with TOF	Generic	✓	✗	✗	✓ flat, isotropic inelastic	✗	✗	

But since it is impossible to read this table – here's where it's at:

<https://confluence.esss.lu.se/display/MCSTAS/McStas+sample+model+functionality-matrix>

... or take a look in school GitHub for at PDF-version.

	available in repo	available in repo	available in repo	not yet available in repo	not yet available in repo	available in repo	available in repo	not yet available in repo	available in repo
17	"4D S'(vec(Q), omega)" Duc Le - soon at ISIS STFC?	Ala Isotropic_Sqw, but with crystal lattice	Elastic and inelastic experiments with crystals	✓	✓	✓	✓	✓	?_L?
18	"Polycrystal" Alberto Cereser + Erik Knudsen, DTU Physics	Engineering-diffraction / imaging oriented multigrain sample	Engineering-diffraction / imaging	✓	✓	✓ (Bragg spots)	✗	✓	✓
19	"Magnetic single crystal" Linda Udby KU, + Erik Knudsen, DTU	Bragg spots from lattice ala Single_crystal plus magnetic lattice. Tabular input (lau)	Single crystal magnetic diffraction	✓	✓	✓ (Bragg spots)	✗	✓	✓ / ?_L?
20	"Reflectometry sample" Jochen Stahn, PSI	Reflectivity-curve sample	Reflectometry	✓	✓	✓ - Reflectivity curve	✗	✗	✗

No labels

Incoherent Sample



Incoherent.comp

Parameters in **boldface** are required; the others are optional.

An incoherent scatterer with various sample shape options

Name	Unit	Description	Default
geometry	str	Name of an Object File Format (OFF) or PLY file for complex geometry. The OFF/PLY file may be generated from XYZ coordinates using qhull/powercrust	0
radius	m	Outer radius of sample in (x,z) plane	0
xwidth	m	Horiz. dimension of sample (bounding box if off file), as a width	0
yheight	m	Vert. dimension of sample (bounding box if off file), as a height. A sphere shape is used when 0 and radius is set	0
zdepth	m	Depth of sample (bounding box if off file)	0
thickness	m	Thickness of hollow sample	0
target_x	-		0
target_y	m	position of target to focus at	0
target_z	-		0
focus_r	m	Radius of disk containing target. Use 0 for full space	0
focus_xw	m	horiz. dimension of a rectangular area	0
focus_yh	m	vert. dimension of a rectangular area	0
focus_aw	deg	horiz. angular dimension of a rectangular area	0
focus_ah	deg	vert. angular dimension of a rectangular area	0
target_index	1	Relative index of component to focus at, e.g. next is +1	0
pack	1	Packing factor	1
p_interact	1	MC Probability for scattering the ray; otherwise transmit	1
f_QE	1	Fraction of quasielastic scattering (rest is elastic)	0
gamma	1	Lorentzian width of quasielastic broadening (HWHM)	0
sigma_abs	barns	Absorption cross section pr. unit cell at 2200 m/s	5.08
sigma_inc	barns	Incoherent scattering cross section pr. unit cell	5.08
Vc	AA ³	Unit cell volume	13.827
concentric	1	Indicate that this component has a hollow geometry and may contain other components. It should then be duplicated after the inside part (only for box, cylinder, sphere)	0
order	-	Limit multiple scattering up to given order	0

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Incoherent Sample



Incoherent.comp

Parameters in **boldface** are required; the others are optional.

An incoherent scatterer with various sample shape options

Name	Unit	Description	Default
geometry	str	Name of an Object File Format (OFF) or PLY file for complex geometry. The OFF/PLY file may be generated from XYZ coordinates using qhull/powercrust	0
radius	m	Outer radius of sample in (x,z) plane	0
xwidth	m	Horiz. dimension of sample (bounding box if off file), as a width	0
yheight	m	Vert. dimension of sample (bounding box if off file), as a height. A sphere shape is used when 0 and radius is set	0
zdepth	m	Depth of sample (bounding box if off file)	0
thickness	m	Thickness of hollow sample	0
target_x	-		0
target_y	m	position of target to focus at	0
target_z	-		0
focus_r	m	Radius of disk containing target. Use 0 for full space	0
focus_xw	m	horiz. dimension of a rectangular area	0
focus_yh	m	vert. dimension of a rectangular area	0
focus_aw	deg	horiz. angular dimension	0
focus_ah	deg	vert. angular dimension	0
target_index	1	Relative index of component	0
pack	1	Packing factor	0
p_interact	1	MC Probability for scattering	0
f QE	1	Fraction of quasielastic scattering	0
gamma	1	Lorentzian width of quasielastic scattering	0
sigma_abs	barns	Absorption cross section	0
sigma_inc	barns	Incoherent scattering cross section	0
Vc	AA^3	Unit cell volume	13.827
concentric	1	Indicate that this component has a hollow geometry and may contain other components. It should then be duplicated after the inside part (only for box, cylinder, sphere)	0
order	-	Limit multiple scattering up to given order	0

p_interact = The probability that a given ray will interact (scatter) with a sample. Does not directly affect the intensity – only the quality of the resulting Monte Carlo estimate.

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Crystalline Samples



From single crystal / crystallites to powder....



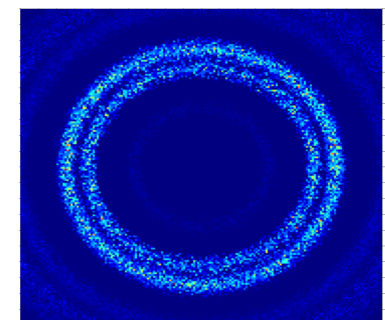
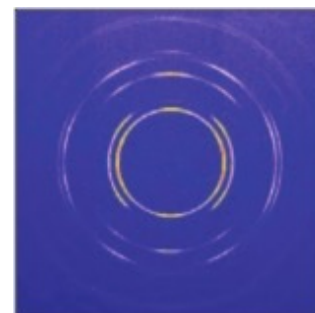
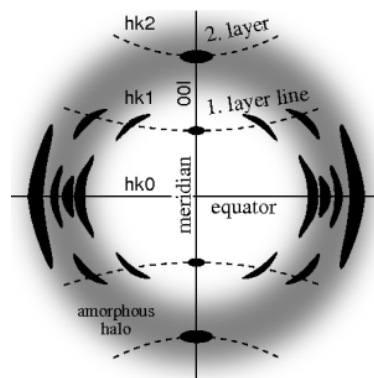
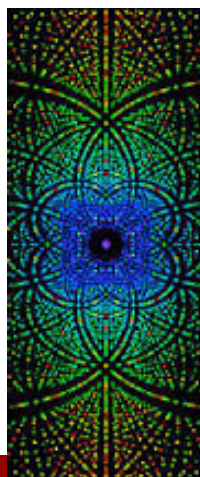
Single crystal



Polycrystal with a little disorder, i.e. a *preferred orientation*, *texture*

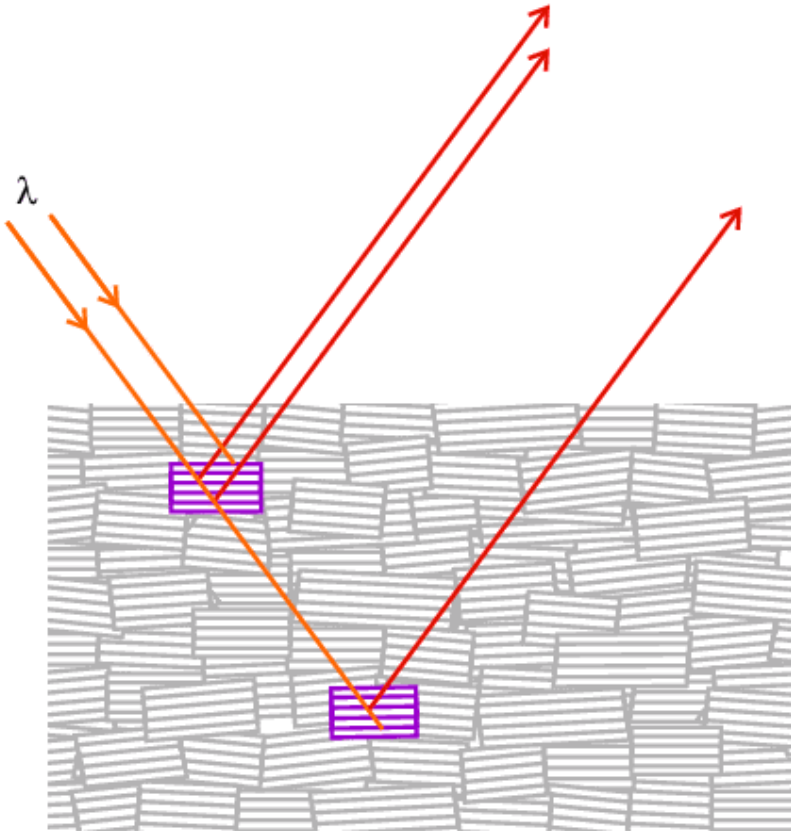


Powder with complete disorder



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Single Crystal model



- *Models ideally imperfect crystal:*
- *Peak broadening is dominated by mosaic
=> not appropriate for highly perfect
crystals.*
- *“Space group symmetries” considered out
of scope.*
 - *Optimized for small unit cells but...*
 - *No assumption about unit cell,
orientation etc.*
 - *Gaussian model for mosaic (see later)*

Courtesy University College London



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Single Crystal



Input parameters

Parameters in **boldface** are required; the others are optional.

Name	Unit	Description	Default
mosaic_AB	arc_minutes, arc_minutes, 1, 1, 1, 1, 1, 1	In Plane mosaic rotation and plane vectors (anisotropic), mosaic_A, mosaic_B, A_h,A_k,A_l, B_h,B_k,B_l. Puts the crystal in the in-plane mosaic state. Vectors A and B define plane in which the crystal roation is defined, and mosaic_A, mosaic_B, denotes the resp. mosaicities (gaussian RMS) with respect to the the two reflections chosen by A and B (Miller indices).	Mosaic_AB_Undefined
reflections	string	File name containing structure factors of reflections. Use empty ("") or NULL for incoherent scattering only	0
geometry	str	Name of an Object File Format (OFF) or PLY file for complex geometry. The OFF/PLY file may be generated from XYZ coordinates using qhull/powercrust	0
xwidth	m	Width of crystal	0
yheight	m	Height of crystal	0
zdepth	no extinction simulated	[m] Depth of crystal	0
radius	m	Outer radius of sample in (x,z) plane	0
delta_d_d	1	Lattice spacing variance, gaussian RMS	1e-4
mosaic	arc minutes	Crystal mosaic (isotropic), gaussian RMS. Puts the crystal in the isotropic mosaic model state, thus disregarding other mosaicity parameters.	-1
mosaic_a	arc minutes	Horizontal (rotation around lattice vector a) mosaic (anisotropic), gaussian RMS. Put the crystal in the anisotropic crystal vector state. I.e. model mosaicity through rotation around the crystal lattice vectors. Has precedence over in-plane mosaic model.	-1
mosaic_b	arc minutes	Vertical (rotation around lattice vector b) mosaic (anisotropic), gaussian RMS.	-1
mosaic_c	arc minutes	Out-of-plane (Rotation around lattice vector c) mosaic (anisotropic), gaussian RMS	-1
recip_cell	1	Choice of direct/reciprocal (0/1) unit cell definition	0
bams	1	Flag to indicate if F ^2 from 'reflections' is in barns or fm^2. barns=1 for laz and isotropic constant elastic scattering (reflections=NULL), barns=0 for lau type files	0
ax	AA or AA^-1	Coordinates of first (direct/recip) unit cell vector	0
ay			0
az			0
bx	AA or AA^-1	Coordinates of second (direct/recip) unit cell vector	0
by			0
bz			0
cx	AA or AA^-1	Coordinates of third (direct/recip) unit cell vector	0
cy			0
cz			0
p_transmit	1	Monte Carlo probability for neutrons to be transmitted without any scattering. Used to improve statistics from weak reflections	-1
sigma_abs	barns	Absorption cross-section per unit cell at 2200 m/s	0
sigma_inc	barns	Incoherent scattering cross-section per unit cell Use -1 to unactivate	0
aa	deg	Unit cell angles alpha, beta and gamma. Then uses norms of vectors a,b and c as lattice parameters	0
bb	deg	Beta angle	0
cc	deg	Gamma angle	0
order	0: all, 1: first, 2: second, ...	[1] Limit multiple scattering up to given order	0
RX	m	Radius of horizontal along X lattice curvature. flat for 0	0
RY	m	Radius of vertical lattice curvature. flat for 0	0
RZ	m	Radius of horizontal along Z lattice curvature. flat for 0	0
powder	1	Flag to indicate powder mode, for simulation of Debye-Scherrer cones via random crystallite orientation. A powder texture can be approximated with 0	0
PG	1	Flag to indicate "Pyrolytic Graphite" mode, only meaningful with choice of Graphite.lau, models PG crystal. A powder texture can be approximated with 0	0

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Single Crystal



Input parameters

Parameters in **boldface** are required; the others are optional.

Name	Unit	Description	Default
mosaic_AB	arc_minutes, arc_minutes, 1, 1, 1, 1, 1	In Plane mosaic rotation and plane vectors (anisotropic), mosaic_A, mosaic_B, A_h,A_k,A_l, B_h,B_k,B_l. Puts the crystal in the in-plane mosaic state. Vectors A and B define plane in which the crystal rotation is defined, and mosaic_A, mosaic_B, denotes the resp. mosaicities (gaussian RMS) with respect to the the two reflections chosen by A and B (Miller indices).	Mosaic_AB_Undefined
reflections	string	File name containing structure factors of reflections. Use empty ("") or NULL for incoherent scattering only	0
geometry	str	Name of an Object File Format (OFF) or PLY file for complex geometry. The OFF/PLY file may be generated from XYZ coordinates using qhull/powercrust	0
xwidth	m	Width of crystal	0
yheight	m	Height of crystal	0
zdepth	no extinction simulated	[m] Depth of crystal	0
radius	m	Outer radius of sample in (x,z) plane	0
delta_d_d	1	Lattice spacing variance, gaussian RMS	1e-4
mosaic	arc minutes	Crystal mosaic (isotropic), gaussian RMS. Puts the crystal in the isotropic mosaic model state, thus disregarding other mosaicity parameters.	-1
mosaic_a	arc minutes	Horizontal (rotation around lattice vector a) mosaic (anisotropic), gaussian RMS. Put the crystal in the anisotropic crystal vector state. I.e. model mosaicity through rotation around the crystal lattice vectors. Has precedence over in-plane mosaic model.	-1
mosaic_b	arc minutes	Vertical (rotation around lattice vector b) mosaic (anisotropic), gaussian RMS.	-1
mosaic_c	arc minutes	Out-of-plane (Rotation around lattice vector c) mosaic (anisotropic), gaussian RMS	-1
recip_cell	1	Choice of direct/reciprocal (0/1) unit cell definition	0
bams	1	Flag to indicate if F ^2 from 'reflections' is in bams or fm^2. bams=1 for laz and isotropic constant elastic scattering (reflections=NULL), bams=0 for lau type files	0
ax	AA or AA^-1	Coordinates of first (direct/recip) unit cell vector	0
ay			0
az			0
bx	AA or AA^-1	Coordinates of second (direct/recip) unit cell vector	0
by			0
bz			0
cx	AA or AA^-1		0
cy			0
cz			0
p_transm	1		-1
sigma_pos	bams		0
sigma_inc	bams	Incoherent scattering cross-section per unit cell Use -1 to unactivate	0
aa	deg	Unit cell angles alpha, beta and gamma. Then uses norms of vectors a,b and c as lattice parameters	0
bb	deg	Beta angle	0
cc	deg	Gamma angle	0
order	0: all, 1: first, 2: second, ...	[1] Limit multiple scattering up to given order	0
RX	m	Radius of horizontal along X lattice curvature. flat for 0	0
RY	m	Radius of vertical lattice curvature. flat for 0	0
RZ	m	Radius of horizontal along Z lattice curvature. flat for 0	0
powder	1	Flag to indicate powder mode, for simulation of Debye-Scherrer cones via random crystallite orientation. A powder texture can be approximated with 0	0
PG	1	Flag to indicate "Pyrolytic Graphite" mode, only meaningful with choice of Graphite.lau, models PG crystal. A powder texture can be approximated with 0	0

Unit cell vectors (ax,ay,az, bx, by,bz, cx,cy, cz). Encodes the crystal orientation relative to the Component's orientation

Single Crystal



Input parameters

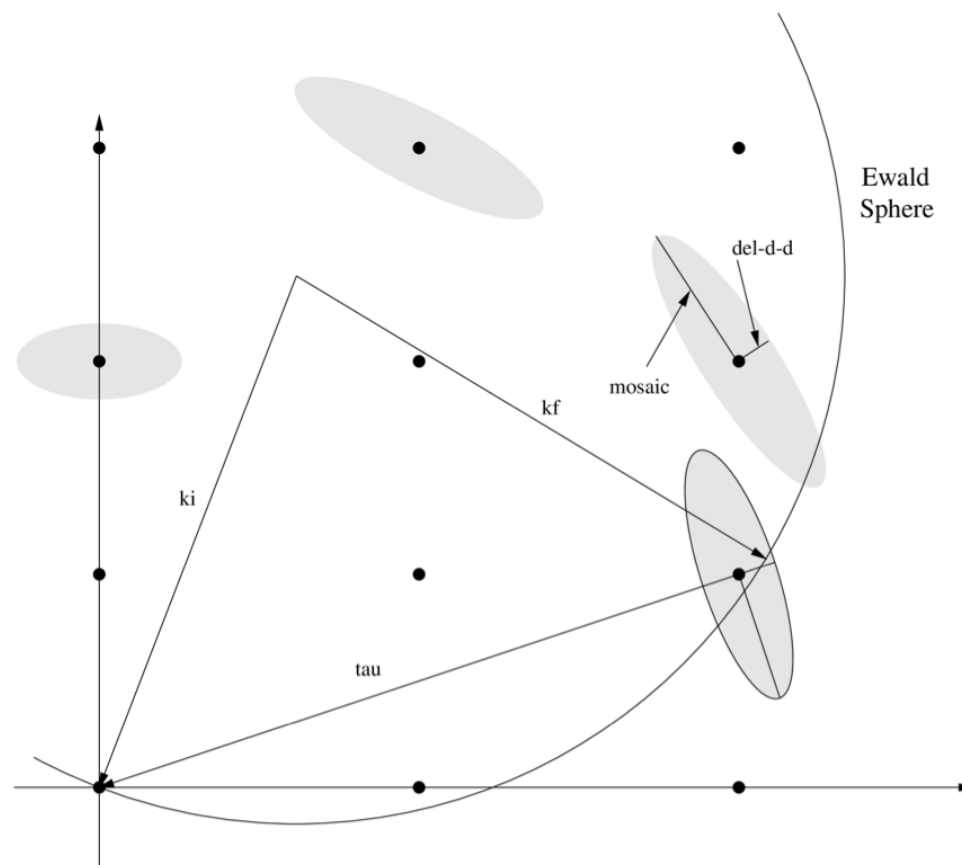
Parameters in **boldface** are required; the others are optional.

Name	Unit	Description	Default
mosaic_AB	arc_minutes, arc_minutes, 1, 1, 1, 1, 1	In Plane mosaic rotation and plane vectors (anisotropic), mosaic_A, mosaic_B, A_h, A_k, A_l, B_h, B_k, B_l. Puts the crystal in the in-plane mosaic state. Vectors A and B define plane in which the crystal roation is defined, and mosaic_A, mosaic_B, denotes the resp. mosaicities (gaussian RMS) with respect to the the two reflections chosen by A and B (Miller indices).	Mosaic_AB_Undefined
reflections	string	File name containing structure factors of reflections. Use empty ("") or NULL for incoherent scattering only	0
g_object	str	Name of an Object File Format (OFF) or PLY file for complex geometry. The OFF/PLY file may be generated from XYZ coordinates using qhull/powercrust	0
xwidth	m	Width of crystal	0
yheight	m	Height of crystal	0
zdepth	[m] no extinction simulated	[m] Depth of crystal	0
radius	m	Outer radius of sample in (x,z) plane	0
laz	1		1e-4
mosaic	arc_minutes		-1
mosaic_a	arc_minutes		-1
mosaic_b	arc_minutes		-1
mosaic_c	arc_minutes		-1
recip_cell	1		0
barns	1	Flag to indicate if $ F ^2$ from 'reflections' is in barns or fm ² . barns=1 for laz and isotropic constant elastic scattering (reflections=NULL), barns=0 for lau type files	0
ax	AA or AA^-1	Coordinates of first (direct/recip) unit cell vector	0
ay			0
az			0
bx	AA or AA^-1	Coordinates of second (direct/recip) unit cell vector	0
by			0
bz			0
cx	AA or AA^-1	Coordinates of third (direct/recip) unit cell vector	0
cy			0
cz			0
p_transmit	1	Monte Carlo probability for neutrons to be transmitted without any scattering. Used to improve statistics from weak reflections	-1
sigma_abs	barns	Absorption cross-section per unit cell at 2200 m/s	0
sigma_inc	barns	Incoherent scattering cross-section per unit cell Use -1 to unactivate	0
aa	deg	Unit cell angles alpha, beta and gamma. Then uses norms of vectors a,b and c as lattice parameters	0
bb	deg	Beta angle	0
cc	deg	Gamma angle	0
order	0: all, 1: first, 2: second, ...	[1] Limit multiple scattering up to given order	0
RX	m	Radius of horizontal along X lattice curvature. flat for 0	0
RY	m	Radius of vertical lattice curvature. flat for 0	0
RZ	m	Radius of horizontal along Z lattice curvature. flat for 0	0
powder	1	Flag to indicate powder mode, for simulation of Debye-Scherrer cones via random crystallite orientation. A powder texture can be approximated with 0	0
PG	1	Flag to indicate "Pyrolytic Graphite" mode, only meaningful with choice of Graphite.lau, models PG crystal. A powder texture can be approximated with 0	0

Mosaicity parameters: mosaic, mosaic_a, mosaic_b, mosaic_c, mosaic_AB

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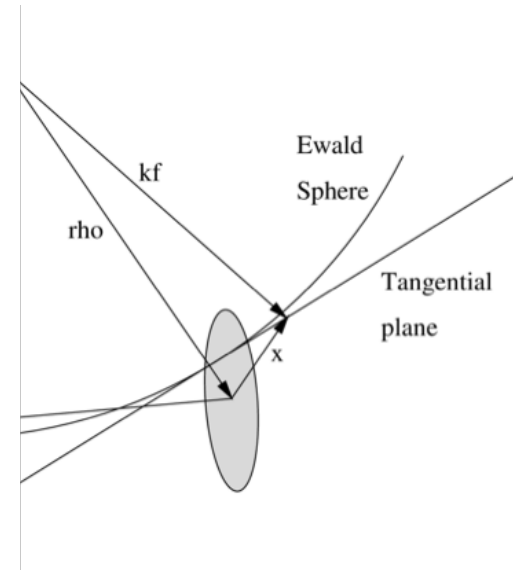
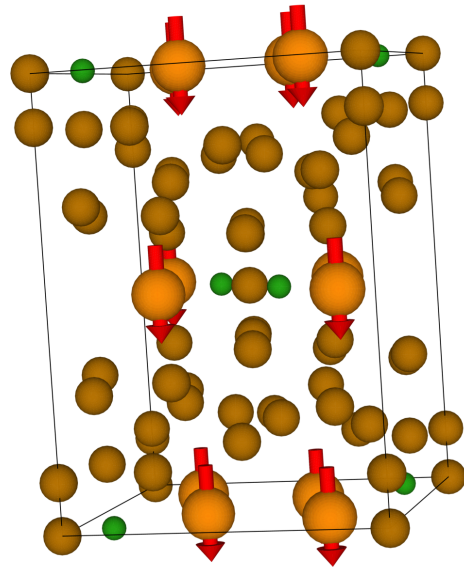
Single Crystal mosaicity



Single Crystal mosaicity

Three models:

- Isotropic Mosaicity = Gaussian distribution of rotation angle, with given standard deviation, around a random vector.
- Anisotropic mosaicity = Gaussian distribution of angles around each of the unit cell axis.

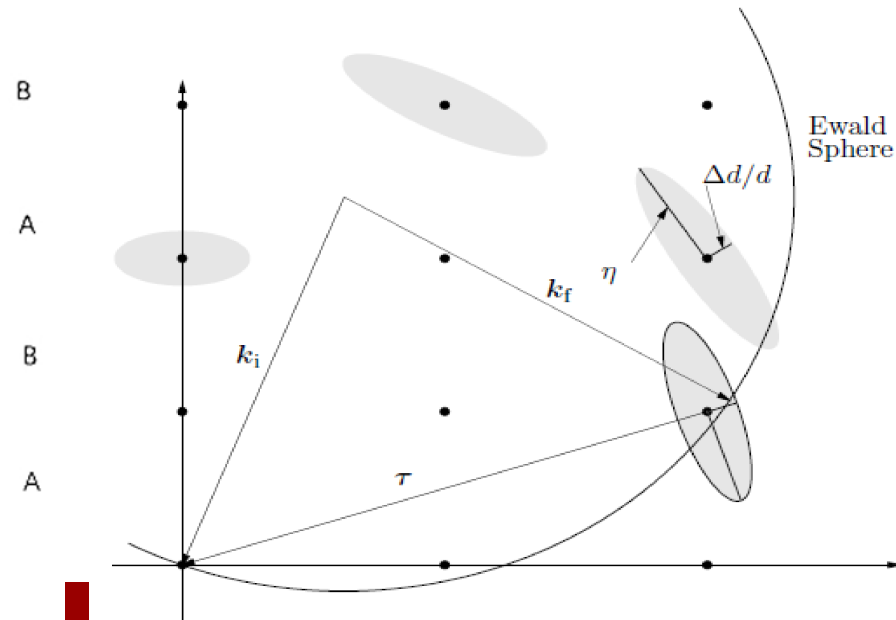
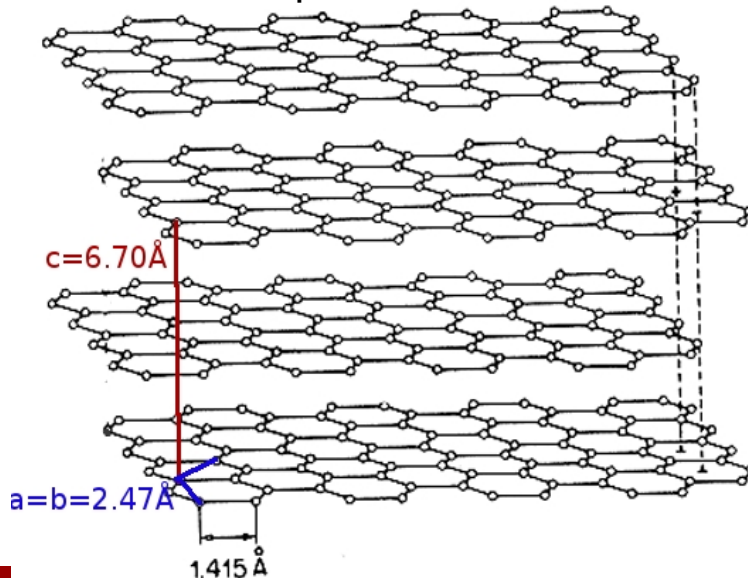


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Single crystal mosaicity

Three models:

- Isotropic Mosaicity = Gaussian distribution of rotation angle, with given standard deviation, around a random vector.
- Anisotropic mosaicity = Gaussian distribution of angles around each of the unit cell axis.
- “Rocking curve mosaic” = Gaussian distr. around axis determined by width of two given Bragg peaks. Specify σ , and hkl for two peaks.



9.2 PowderN



Among the most common materials measured with neutron scattering techniques are powders, which are composed of a large number of tiny single crystals. The scattering intensity, by means of constructive interference and averaging over all crystallites, appears as well defined rings around the out-going direct beam direction. These rings are also present in liquids, but are much smoother, as we shall see in next section.

Let us consider a material of density ρ made of N scattering units each with a unit cell volume V_0 arranged with lattice spacings $d_Q = 2\pi/Q$ associated to structure factors $F(Q)$ with equivalent reflection multiplicities j_Q . These structure factors characterize the efficiency of the reflection with momentum exchange Q . Following Squires [14] the scattering probability for an incoming neutron with wavelength λ penetrating along a distance x into the material is about $1 - \exp(-\rho\sigma_{\text{cone}} x)$ where σ_{cone} is the so-called coherent elastic cross-section of the ring,

$$\sigma_{\text{cone}} = \frac{N \pi \lambda^2 j_Q |F(Q)|^2}{V_0 Q} .$$

This relation is only valid under certain conditions, among which $d_Q > \lambda/2$, from the Bragg law. As the possible lattice spacings d in the material can not exceed a maximum value (for instance the inter-atomic distance), it appears that as the neutron wavelength increases, the number of visible rings in the diffractogram will decrease, until no more scattering is possible above the so-called *Bragg edge*, where materials become transparent to neutrons (except for absorption and incoherent scattering). This is why most of the diffractometers use thermal and hot neutrons. Cold neutrons can only scatter on large distance arrangements in materials, *e.g.* in larger molecules and proteins.

PowderN inputs



Input parameters

Parameters in **boldface** are required; the others are optional.

Name	Unit	Description	Default
format	no quotes	Name of the format, or list of column indexes (see Description).	Undefined
reflections			"NULL"
geometry	str	Name of an Object File Format (OFF) or PLY file for complex geometry. The OFF/PLY file may be generated from XYZ coordinates using qhull/powercrust	"NULL"
radius	m	Outer radius of sample in (x,z) plane	0
yheight	m	Height of sample y direction	0
xwidth	m	Horiz. dimension of sample, as a width	0
zdepth	m	Depth of box sample	0
thickness			0
pack	1	Packing factor	1
Vc	AA ³	Volume of unit cell=nb atoms per cell/density of atoms	0
sigma_abs	barns	Absorption cross section per unit cell at 2200 m/s. Use a negative value to unactivate it	0
sigma_inc	barns	Incoherent cross section per unit cell. Use a negative value to unactivate it	0
delta_d_d	0/1	Global relative delta_d/d broadening when the 'w' column is not available. Use 0 if ideal.	0
p_inc	1	Fraction of incoherently scattered neutron rays	0.1
p_transmit	1	Fraction of transmitted (only attenuated) neutron rays	0.1
DW	1	Global Debye-Waller factor when the 'DW' column is not available. Use 1 if included in F2	0
nb_atoms	1	Number of sub-unit per unit cell, that is ratio of sigma for chemical formula to sigma per unit cell	1
d_phi	deg	Angle corresponding to the vertical angular range to focus to, e.g. detector height. 0 for no focusing	0
p_interact	1	Fraction of events interacting with sample, e.g. 1-p_transmit-p_inc	0
concentric	only for box, cylinder, sphere	[1] Indicate that this component has a hollow geometry and may contain other components. It should then be duplicated after the inside part	0
density	g/cm ³	Density of material. rho=density/weight/1e24*N_A.	0
weight	g/mol	Atomic/molecular weight of material	0
barns	1	Flag to indicate if F ^2 from 'reflections' is in barns or fm^2 (barns=1 for laz, barns=0 for lau type files).	1
Strain	ppm	Global relative delta_d/d shift when the 'Strain' column is not available. Use 0 if ideal.	0
focus_flip	1	Controls the sense of d_phi. If 0 d_phi is measured against the xz-plane. If !=0 d_phi is measured against zy-plane.	0

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PowderN inputs



Input parameters

Parameters in **boldface** are required; the others are optional.

Name	Unit	Description	Default
format	no quotes	Name of the format, or list of column indexes (see Description).	Undefined
reflections			"NULL"
geometry	str	Name of an Object File Format (OFF) or PLY file for complex geometry. The OFF/PLY file may be generated from XYZ coordinates using qhull/powercrust	"NULL"
radius	m	Outer radius of sample in (x,z) plane	0
yheight	m	Height of sample y direction	0
xwidth	m	Horiz. dimension of sample, as a width	0
zdepth	m	Depth of box sample	0
thickness			0
pack	1	Packing factor	1
Vc	AA^3	Volume of unit cell=nb atoms per cell/density of atoms	0
sigma_abs	barns	Absorption cross section per unit cell at 2200 m/s. Use a negative value to unactivate it	0
sigma_inc	barns	Incoherent cross section per unit cell. Use a negative value to unactivate it	0
delta_d_d	0/1	When the 'w' column is not available. Use 0 if ideal.	0
p_inc	1	Fraction of transmitted (only attenuated) neutron rays	0.1
p_transmit	1	Fraction of transmitted (only attenuated) neutron rays	0.1
DW	1	Global Debye-Waller factor when the 'DW' column is not available. Use 1 if included in F2	0
nb_atoms	1	Number of sub-unit per unit cell, that is ratio of sigma for chemical formula to sigma per unit cell	1
d_phi	deg	Angle corresponding to the vertical angular range to focus to, e.g. detector height. 0 for no focusing	0
p_interact	1	Fraction of events interacting with sample, e.g. 1-p_transmit-p_inc	0
concentric	only for box, cylinder, sphere	[1] Indicate that this component has a hollow geometry and may contain other components. It should then be duplicated after the inside part	0
density	g/cm^3	Density of material. rho=density/weight/1e24*N_A.	0
weight	g/mol	Atomic/molecular weight of material	0
barns	1	Flag to indicate if F ^2 from 'reflections' is in barns or fm^2 (barns=1 for laz, barns=0 for lau type files).	1
Strain	ppm	Global relative delta_d/d shift when the 'Strain' column is not available. Use 0 if ideal.	0
focus_flip	1	Controls the sense of d_phi. If 0 d_phi is measured against the xz-plane. If !=0 d_phi is measured against zy-plane.	0

Sampling parameters

Concentric = Hollow
Powder [cylinder]

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PowderN inputs



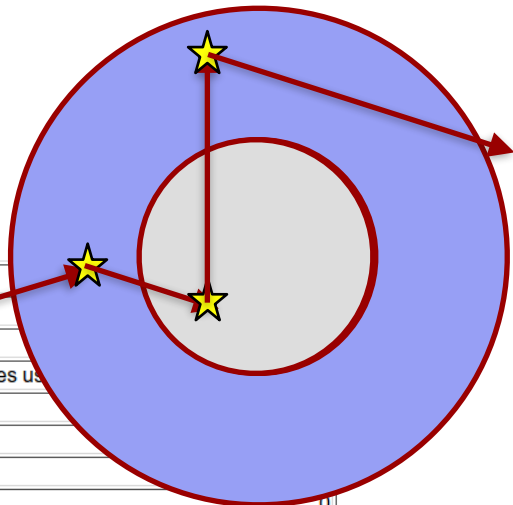
Input parameters

Parameters in **boldface** are required; the others are optional.

Name	Unit	Description	
format	no quotes	Name of the format, or list of column indexes (see Description).	
reflections			
geometry	str	Name of an Object File Format (OFF) or PLY file for complex geometry. The OFF/PLY file may be generated from XYZ coordinates using the 'w' column.	
radius	m	Outer radius of sample in (x,z) plane	
yheight	m	Height of sample y direction	
xwidth	m	Horiz. dimension of sample, as a width	
zdepth	m	Depth of box sample	0
thickness			0
pack	1	Packing factor	1
Vc	AA^3	Volume of unit cell=nb atoms per cell/density of atoms	0
sigma_abs	barns	Absorption cross section per unit cell at 2200 m/s. Use a negative value to unactivate it	0
sigma_inc	barns	Incoherent cross section per unit cell. Use a negative value to unactivate it	0
delta_d_d	0/1	On the 'w' column is not available. Use 0 if ideal.	0
p_inc	1	Days	0.1
p_transmit	1	Fraction of transmitted (only attenuated) neutron rays	0.1
DW	1	Global Debye-Waller factor when the 'DW' column is not available. Use 1 if included in F2	0
nb_atoms	1	Number of sub-unit per unit cell, that is ratio of sigma for chemical formula to sigma per unit cell	1
d_phi	deg	Angle corresponding to the vertical angular range to focus to, e.g. detector height. 0 for no focusing	0
p_interact	1	Fraction of events interacting with sample, e.g. 1-p_transmit-p_inc	0
concentric	only for box, cylinder, sphere	[1] Indicate that this component has a hollow geometry and may contain other components. It should then be duplicated after the inside part	0
density	g/cm^3	Density of material. rho=density/weight/1e24*N_A.	0
weight	g/mol	Atomic/molecular weight of material	0
barns	1	Flag to indicate if F ^2 from 'reflections' is in barns or fm^2 (barns=1 for laz, barns=0 for lau type files).	1
Strain	ppm	Global relative delta_d/d shift when the 'Strain' column is not available. Use 0 if ideal.	0
focus_flip	1	Controls the sense of d_phi. If 0 d_phi is measured against the xz-plane. If !=0 d_phi is measured against zy-plane.	0

Sampling parameters

Concentric = Hollow Powder [cylinder]



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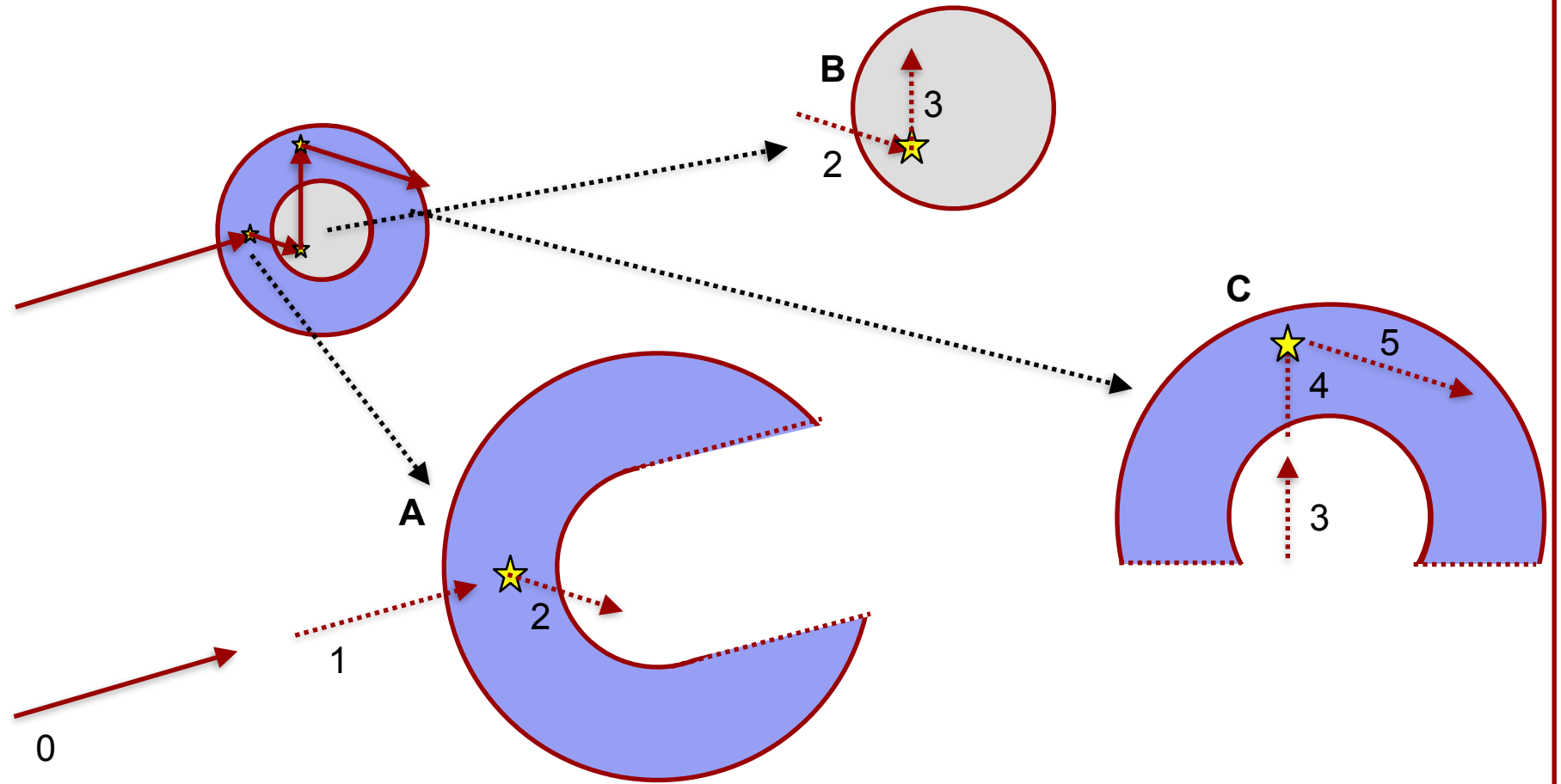
Powder



Input parameters

Parameters in **boldface** are re

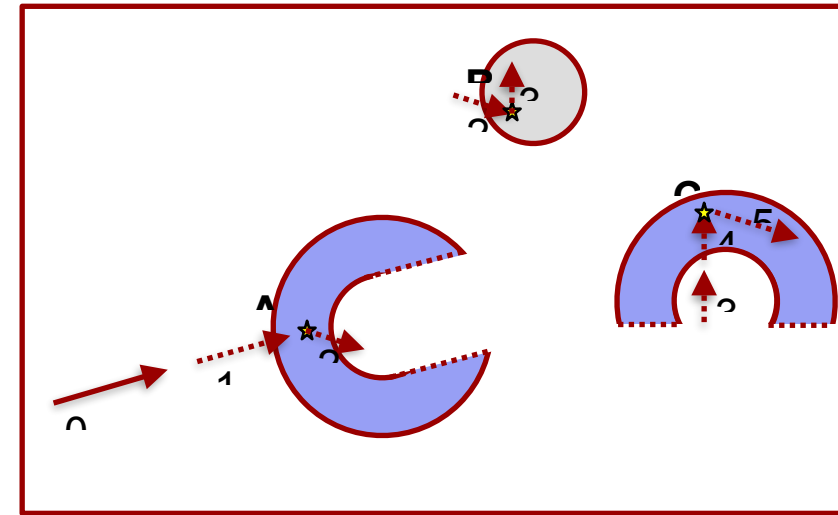
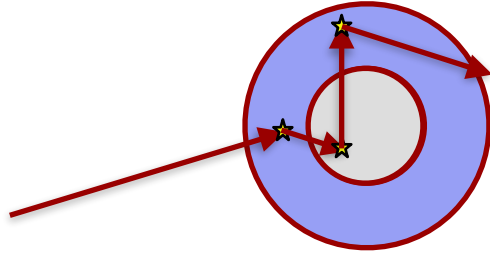
Name	Unit
format	no quotes
reflections	
geometry	str
radius	m
yheight	m
xwidth	m
zdepth	m
thickness	
pack	1
Vc	AA^3
sigma_abs	barns
sigma_inc	barns
d_nk_d_0	0/1
p_inc	1
p_transmit	1
DW	1
nb_stems	1
d_phi	deg
p_interact	1
concentric	only for box, cylind
density	g/cm^3
weight	g/mol
barns	1
Strain	ppm
focus_flip	1



Concentric = Hollow
Powder [cylinder]

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PowderN concentric in code



```
COMPONENT Al = PowderN(
    reflections="Al.laz", radius=0.02, yheight=0.05, thickness=0.005,
    p_transmit=0.8, concentric=1)
AT (0, 0, 0) RELATIVE PREVIOUS
```

```
COMPONENT al2co3 = PowderN(
    reflections=Al2CO3-laz, radius=0.005, yheight=0.05,
    p_inc=0.15, p_transmit=0.5)
AT (0, 0, 0) RELATIVE PREVIOUS
```

```
COMPONENT Al_2 = COPY(Al) (concentric=0)
AT (0,0,0) RELATIVE PREVIOUS
```

PowderN inputs



Input parameters

Parameters in **boldface** are required; the others are optional.

Name	Unit	Description	Default
format	no quotes	Name of the format, or list of column indexes (see Description).	Undefined
reflections			"NULL"
geometry	str	Name of an Object File Format (OFF) or PLY file for complex geometry. The OFF/PLY file may be generated from XYZ coordinates using qhull/powercrust	"NULL"
radius	m	Outer radius of sample in (x,z) plane	0
yheight	m	Height of sample y direction	0
xwidth	m	Horiz. dimension of sample, as a width	0
zdepth	m	Depth of box sample	0
thickness			0
pack	1	Packing factor	1
Vc	AA^3	Volume of unit cell=nb atoms per cell/density of atoms	0
sigma_abs	barns	Absorption cross section per unit cell at 2200 m/s. Use a negative value to unactivate it	0
sigma_inc	barns	Incoherent cross section per unit cell. Use a negative value to unactivate it	0
delta_d_d	0/1	Global relative delta_d/d broadening when the 'w' column is not available. Use 0 if ideal.	0
p_inc	1	Fraction of incoherently scattered neutron rays	0.1
p_transmit	1	Fraction of transmitted (only attenuated) neutron rays	0.1
DW	1	Global Debye-Waller factor when the 'DW' column is not available. Use 1 if included in F2	0
ph_stems	1	Number of sub-unit per unit cell, that is ratio of sigma for chemical formula to sigma per unit cell	1
d_phi	deg	Angle corresponding to the vertical angular range to focus to, e.g. detector height. 0 for no focusing	0
p_interact	1	Fraction of events interacting with sample, e.g. 1-p_transmit-p_inc	0
concentric	only for box, cylinder, sphere	[1] Indicate that this component has a hollow geometry and may contain other components. It should then be duplicated after the inside part	0
density	g/cm^3	Density of material. rho=density/weight/1e24*N_A.	0
weight	g/mol		0
barns	1	s or fm^2 (barns=1 for laz, barns=0 for lau type files).	1
strain	ppm	column is not available. Use 0 if ideal.	0
focus_flip	1	d against the xz-plane. If !=0 d_phi is measured against zy-plane.	0

Only scatter into a part of
4π

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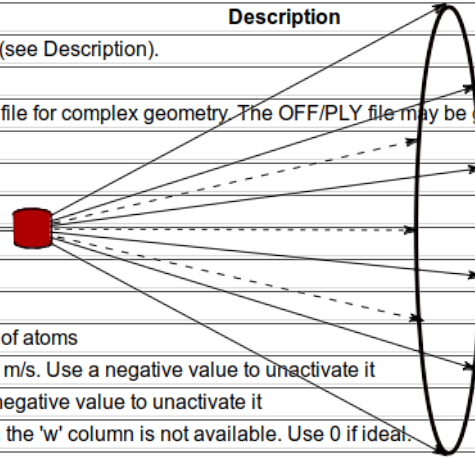
PowderN inputs



Input parameters

Parameters in **boldface** are required; the others are optional.

Name	Unit	Description	Default
format	no quotes	Name of the format, or list of column indexes (see Description).	Undefined
reflections			"NULL"
geometry	str	Name of an Object File Format (OFF) or PLY file for complex geometry. The OFF/PLY file may be generated from XYZ coordinates using qhull/powercrust.	"NULL"
radius	m	Outer radius of sample in (x,z) plane	0
yheight	m	Height of sample y direction	0
xwidth	m	Horiz. dimension of sample, as a width	0
zdepth	m	Depth of box sample	0
thickness			0
pack	1	Packing factor	1
Vc	AA^3	Volume of unit cell=nb atoms per cell/density of atoms	0
sigma_abs	barns	Absorption cross section per unit cell at 2200 m/s. Use a negative value to unactivate it	0
sigma_inc	barns	Incoherent cross section per unit cell. Use a negative value to unactivate it	0
delta_d_d	0/1	Global relative delta_d/d broadening when the 'w' column is not available. Use 0 if ideal.	0
p_inc	1	Fraction of incoherently scattered neutron rays	0.1
p_transmit	1	Fraction of transmitted (only attenuated) neutron rays	0.1
DW	1	Global Debye-Waller factor when the 'DW' column is not available. Use 1 if included in F2	0
ph_stems	1	Number of sub-unit per unit cell, that is ratio of sigma for chemical formula to sigma per unit cell	1
d_phi	deg	Angle corresponding to the vertical angular range to focus to, e.g. detector height. 0 for no focusing	0
p_interact	1	Fraction of events interacting with sample, e.g. 1-p_transmit-p_inc	0
concentric	only for box, cylinder, sphere	[1] Indicate that this component has a hollow geometry and may contain other components. It should then be duplicated after the inside part	0
density	g/cm^3	Density of material. rho=density/weight/1e24*N_A.	0
weight	g/mol		0
barns	1	or fm^2 (barns=1 for laz, barns=0 for lau type files).	1
strain	ppm	column is not available. Use 0 if ideal.	0
focus_flip	1	d against the xz-plane. If !=0 d_phi is measured against zy-plane.	0



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PowderN inputs



Input parameters

Parameters in **boldface** are required; the others are optional.

Name	Unit	Description	Default
format	no quotes	Name of the format, or list of column indexes (see Description).	Undefined
reflections			"NULL"
geometry	str	Name of an Object File Format (OFF) or PLY file for complex geometry. The OFF/PLY file may be generated from XYZ coordinates using qhull/powercrust	"NULL"
radius	m	Outer radius of sample in (x,z) plane	0
yheight	m	Height of sample y direction	0
xwidth	m	<u>Horiz. dimension of sample, as a width</u>	0
zdepth	m	Depth of box sample	0
thickness			0
pack	1	Packing factor	1
Vc	AA^3	Volume of unit cell=nb atoms per cell/density of atoms	0
sigma_abs	barns	Absorption cross section per unit cell at 2200 m/s. Use a negative value to unactivate it	0
sigma_inc	barns	Incoherent cross section per unit cell. Use a negative value to unactivate it	0
delta_d_d	0/1	Global relative <u>delta_d_d/d</u> broadening when the 'w' column is not available. Use 0 if ideal.	0
p_inc	1	Fraction of incoherently scattered neutron rays	0.1
p_transmit	1	Fraction of transmitted (only attenuated) neutron rays	0.1
DW	1	Global Debye-Waller factor when the 'DW' column is not available. Use 1 if included in F2	0
ph_stems	1	Number of sub-unit per unit cell, that is ratio of sigma for chemical formula to sigma per unit cell	1
d_phi	deg	Angle corresponding to the vertical angular range to focus to, e.g. detector height. 0 for no focusing	0
p_interact	1	Fraction of events interacting with sample, e.g. 1-p_transmit-p_inc	0
concentric	only for box, cylinder, sphere	[1] Indicate that this component has a hollow geometry and may contain other components. It should then be duplicated after the inside part	0
density	g/cm^3	Density of material. rho=density/weight/1e24*N_A.	0
weight	g/mol		0
barns	1	s or fm^2 (barns=1 for laz, barns=0 for lau type files).	1
strain	ppm	column is not available. Use 0 if ideal.	0
focus_flip	1	d against the xz-plane. If !=0 d_phi is measured against zy-plane.	0

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EUROPEAN
SPALL
SOUR

TITLE *Aluminum-Al-[FM3-M] Miller, H.P.jr.;DuMond, J.W.M.[1942] at 298 K

CELL 4.049320 4.049320 4.049320 90.000000 90.000000 90.000000

SPCGRP F M 3 M CUBIC STRUCTURE

ATOM AL 1 0.000000 0.000000 0.000000

SCATTERING FACTOR COEFFICIENTS: AL F= 0.345 CM-12

Reference: Physical Review (1940) 57, 198-206

#

Physical parameters:

sigma_coh 1.495 coherent scattering cross section (single atom) in [barn]

sigma_inc 0.0082 incoherent scattering cross section (single atom) in [barn]

sigma_abs 0.231 absorption scattering cross section (single atom) in [barn]

density 2.70 in [g/cm^3]

weight 26.98 in [g/mol] (single atom)

multiplicity 4 in [atoms/unit cell]

Vc 66.4 volume of unit cell in [A^3]

v_sound 5100 in [m/s]

v_sound_l 6420 velocity of longitudinal sound in [m/s]

v_sound_t 3040 velocity of transversal sound in [m/s]

T_m 933.5 melting temperature in [K]

T_b 2792.2 boiling temperature in [K]

At_number 13 atomic number Z

lattice_a 4.04932 lattice parameter a in [Angs]

#

Format parameters: Crystallographica format

column_j 4 multiplicity 'j'

column_d 5 d-spacing 'd' in [Angs]

column_f 5 scattering factor |F|^2 in [fm^2]



Lau datafiles

header

+

reflection list

Can be used with
Single_crystal, PowderN,
Isotropic_Sqw

Input for the PowderN



```
# TITLE *Corundum-Al2O3-[R3-CH] Graafsma, H.;Souhassou, M.;Harkem[1998] [corundum sapphire:blue, ruby:red]

# CELL 4.757000 4.757000 12.987700 90.000000 90.000000 120.000000

# SPCGRP R -3 C  TRIGONAL STRUCTURE

# ATOM AL 1 0.000000 0.000000 0.352110

# ATOM O  1 0.306260 0.306260 0.250000

# SCATTERING FACTOR  COEFFICIENTS: AL   F= 0.345 CM-12 ; O    F= 0.581 CM-12

# Reference: Acta Crystallographica B (1998) 54, 193-195

#

# Physical parameters:

# sigma_coh 15.683  coherent scattering cross section for Al2O3 in [barn]

# sigma_inc 0.0188  incoherent scattering cross section for Al2O3 in [barn]

# sigma_abs 0.4625  absorption scattering cross section for Al2O3 in [barn]

# density  4.05   in [g/cm^3]

# weight  101.96  in [g/mol] for Al2O3

# multiplicity 6   in [Al2O3/unit cell]

# Vc      254.52  volume of unit cell in [A^3]

# T_m     2273    melting temperature in [K]

# T_b     3773    boiling temperature in [K]

# lattice_a 4.757  lattice parameter a in [Angs]

# lattice_c 12.9877 lattice parameter c in [Angs]

# lattice_cc 120   lattice angle gamma in [deg]

#

# Format parameters: Lazy format <http://icsd.ill.fr>

# column_j 17 multiplicity 'j'

# column_d 6  d-spacing 'd' in [Angs]

# column_F 13 norm of scattering factor |F| in [barn]

# column_h 1
```

Laz + Lau datafiles

header

+

reflection list

Can be used with PowderN,
Isotropic_Sqw

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What's already there?

Single Crystals

Al.lau
Al2O3_sapphire.lau
B4C.lau
BCC_fake.lau
BeO.lau
C60.lau
C_diamond.lau
C_graphite.lau
CaF2.lau
Ge.lau
In.lau
PPase_D_P1.lau
Rubredoxin.lau
SiO2_cristobalite.lau
SiO2_quartza.lau
SiO2_quartzb.lau
TiO2_rutile.lau
YBaCuO.lau
adrenaline.lau
aspirin.lau
leucine.lau

Powders

Ag.laz
Al.laz
Al2O3_sapphire.laz
Au.laz
B4C.laz
Ba.laz
Be.laz
BeO.laz
Bi.laz
C_diamond.laz
C_graphite.laz
Cd.laz
Cr.laz
Cs.laz
Cu.laz
Cu2MnAl.laz
Fe.laz
Fe_Gamma.laz
Ga.laz
Gd.laz
Ge.laz
H2O_ice_1h.laz
He4_hcp.laz
Hg.laz
I2.laz
K.laz
Li.laz
LiF.laz
Mo.laz
Na2Ca3Al2F14.laz
Nb.laz
Ni.laz
Pb.laz
Pd.laz
Pt.laz
Rb.laz
Se_alpha.laz
Se_beta.laz
Si.laz
SiO2_quartza.laz
SiO2_quartzb.laz
Sn_alpha.laz
Sn_beta.laz
Ti.laz
Tl.laz
UO2.laz
V.laz
Y2O3.laz
Y3Fe5O12_YIG.laz
Zn.laz
Zr.laz
duplex.laz

Where to get these files...

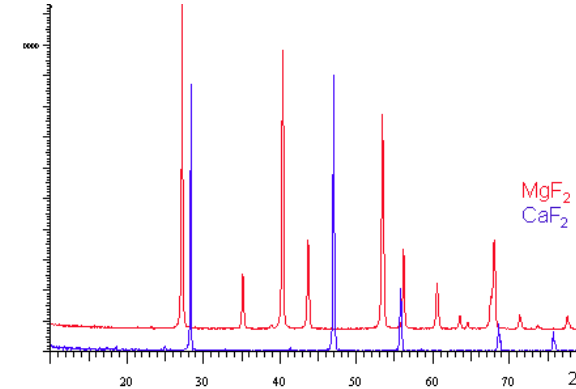


- | \$MCSTAS/data
 - | Windows: c:\mcstas-2.5\lib\data
 - | Linux: /usr/(local)/share/mcstas/2.5/data
 - | OS X: /Applications/McStas-2.5/Contents/Resources/mcstas/2.5/data
- | - Or make your own via
 - | Finding a CIF file for the given structure
 - | e.g. from
ICSD <http://icsd.fiz-karlsruhe.de> or
COD <<http://crystallography.net>>
 - | [Build one using VESTA](#)
 - | Process it using
 - | cif2hkl which is available in your McStas installation

McStas: structure: powder

•For powders, the pure elastic diffraction is easy to prepare: LAZ

- PowderN
- Single_crystal(powder=1, ...)
- Isotropic_Sqw



CIF file

FullProf/ShelX

Terminal:

```
cif2hkl --powder file.cif
```

McStas Laz file

McStas: structure: SX

•For single crystals, the pure elastic diffraction is easy to prepare: LAU

- PowderN
- Single_crystal
- Isotropic_Sqw



CIF file

FullProf/ShelX

Terminal:

```
cif2hkl --xtal file.cif
```

McStas Lau file

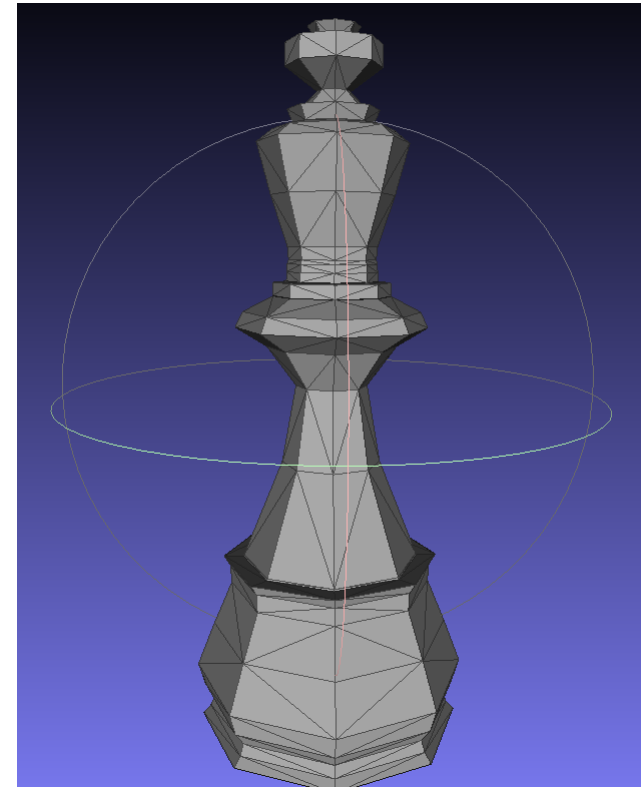
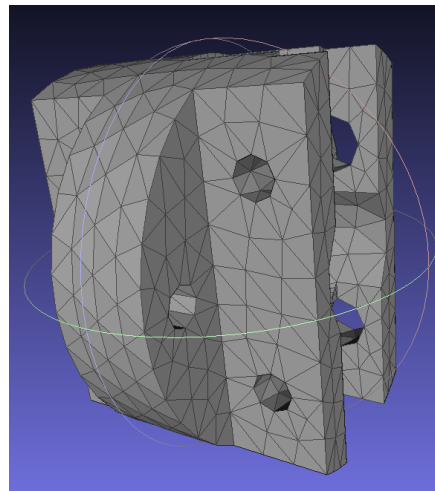
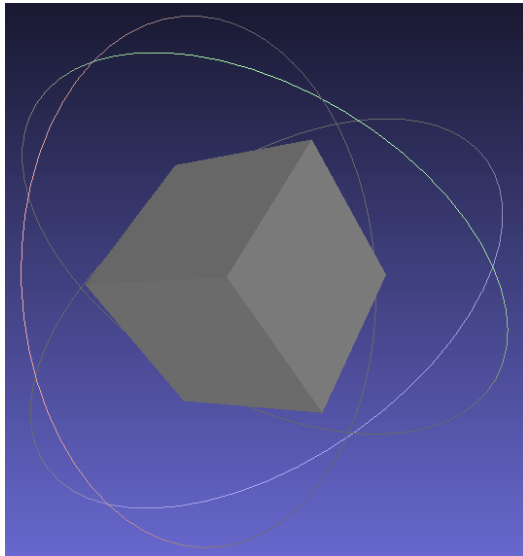
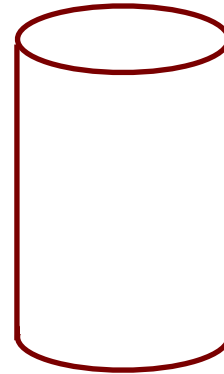
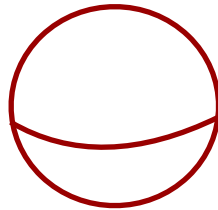
McStas uses a number of input data formats, all text based.

- **Lau** (Single_crystal, PowderN, Isotropic_Sqw)
- **Laz** (PowderN, Isotropic_Sqw)
- **Sqw** (Isotropic_Sqw)
- **qSq** (Isotropic_Sqw) – mostly for liquids
- **nxs** (Sample_nxs) – this is *not* a NeXus/HDF
- **PDB** (SANSPDB)
- **ncmat** (NCrystal)

OFF/PLY any geometry



An additional complex geometry enables to use any point set to describe the material volume (*geomview* OFF file).



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